2. Reaction with metals

a) Wurtz Fittig reaction

Halo arenes reacts with halo alkanes when heated with sodium in ether solution to form alkyl benzene. This reaction is called Wurtz Fittig reaction.

$$C_6H_5Cl + 2Na + ClC_2H_5 \xrightarrow[\Delta]{\text{Ether}} C_6H_5C_2H_5 + 2NaCl$$

Chlorobenzene chloroethane ethyl benzene

b) Fittig reaction

Haloarenes react with sodium metal in dry ether, two aryl groups combine to give biaryl products. This reaction is called Fittig reaction

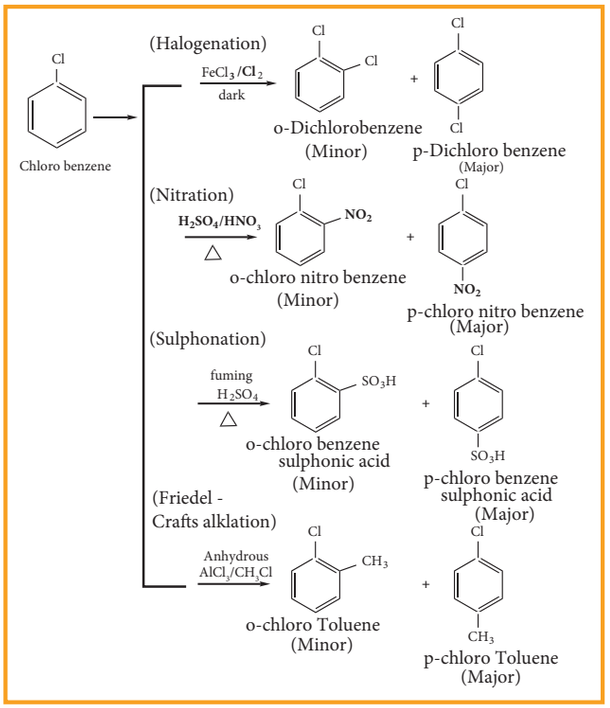
$$C_6H_5Cl + 2Na + Cl\text{-}C_6H_5 \xrightarrow[\Delta]{\text{Ether}} C_6H_5 - C_6H_5 + 2NaCl$$

Chlorobenzene Biphenyl

B) Reaction involving aromatic ring

3. Electrophilic substitution reaction

Haloarenes undergo aromatic electrophilic substitution reactions. The rate of eleclophilic substitution of halobenzene is lower than that of benzene. halogen is deactivating due to - I effect of halogen. The lone pair of electrons on the chlorine involves in resonance with the ring. It increases the electron density at ortho and para position (refer figure no 14.1). The halogen attached to the benzine ring with draw electron and thereby and hence the halogen which is attached to the benzene directs the incoming, electrophile either to ortho or to para position in electrophilie substitution reaction



Toluene

4) Reduction

Haloarenes on reduction with Ni-Al alloy in the presence of NaOH gives corresponding arenes.

$$C_6H_5Cl + 2(H) \xrightarrow[NaOH]{\text{Ni-Al}} C_6H_6 + HCl$$

Chloro benzene Benzene

5) Formation of Grignard reagent

Haloarenes reacts with magnesium to form Grignard reagent in tetra hydrofuran (THF).

$$C_6H_5Cl + Mg \xrightarrow{THF} C_6H_5MgCl$$

Chloro benzene Phenyl magnesium chloride

14.5.6 Uses of Chloro benzene

i) Chloro benzene is used in the manufacture of pesticides like DDT

ii) It is used as high boiling solvent in organic synthesis.

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